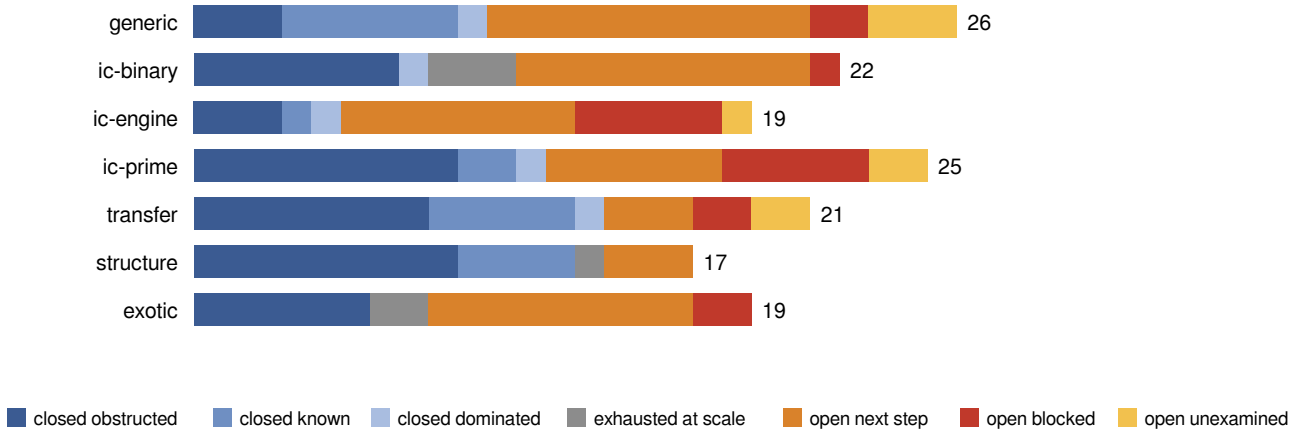
ECDLP lead atlas: review of 2026-10-29 (prime-field IC delta)

Question. What changed in the four repositories between the first full review (2026-10-24, `docs/reports/ecdlp-lead-atlas-20261024/`) and 2026-10-29? Which lead dispositions did those changes move?

Scope and method. One review agent ran in the `prime-ic-prep` workflow of 2026-10-29, followed by an independent cross-track reviewer. It ran `python3 tools/lead_atlas.py --delta` and folded every listed path into its owning lead: 227 watched paths in crypto-autoresearcher, 93 in crypto and 77 in cryptanalysis. It mapped the unmapped ECDLP ledger records with `--coverage`, ran `--check` until it passed, and stamped the review `prime-ic-prep-20261029`. Prime-field index calculus (`ic-prime`, `ic-engine`) had priority, because that was the session's question. Companion program: crypto `research/prime_ic_program_20261029/`.

Status of each statement. Dispositions are derived from committed evidence and cite it. They are not new measurements, and they change no hypothesis, experiment or goal status. Numbers are measured values copied from the cited records.

ECDLP lead atlas, 2026-10-29: 149 leads by family and disposition



Result

- **149 leads**: 74 closed with a named obstruction (45 obstructed, 18 known, 5 dominated, 6 exhausted at scale), 50 open with a decided next step, 17 blocked, 8 unexamined.
- **Coverage 2092/2092.** Every ECDLP-area ledger record maps to a lead; this review added 44 rows to `coverage.tsv`.
- **Two disposition changes**, both in `ic-prime`:
  - **New** `LEAD-ICP-pkm-tower-algebraic-oracle` (open_blocked). It covers PKM tower and APS partitioned factor bases solved algebraically over F_p, measured by first and last fall degree. No lead covered it before, although it is the only prime-field route that meets the owner's 2026-10-24 algebraic criterion. Blocked by the fall-degree relaunch route (harness host), the stranded Mac runs, and memory for m = 4 at N = 20.
  - `LEAD-ICP-coppersmith-smallroot-semaev`: open_next_step to **closed_obstructed** (proof, scoped). Any lattice built from (p, f)^t on one S_3 block with equal per-variable bounds reaches at most 0.3293 (three unknowns) or 0.3536 (two). The optimiser found nothing above 1/2 with unequal bounds. The demand is 8/9 (crypto `research/smallroot_optimal_reach_20261026`). This **supersedes next step 6 of the 2026-10-24 report** ("optimal Coppersmith shift set"): that calculation is done and closed the lead. `revisit_when` names joint three-block lattices, unequal bounds above 1/2, rank-r GAP bases and non-Coppersmith solvers. The unequal-bound value 1/2 is numerical, not proved.
- **Folded without a disposition change** (prime-field IC):

  - `LEAD-ICP-prime-ic-constant-engineering`: `best_measured` is now the 2026-10-29 20-bit factor-base change, N 32 → 45, Σ S_sel/Σ S_inc 0.818 [0.773, 0.866] at w = 0 and 0.726 [0.677, 0.778] at w = 1 (crypto `research/prime_base_oracle_tuning_20261029`). Its next action is that row's second-host replay, then the hash-table floor.
  - `LEAD-ICP-residual-partial-relations`: on-main evidence from crypto `ic_prime_residual_lp_20261026` and `ic_planner_exhaustive_20261026`. The residual collector is a birthday search; it is generic on the curve-free null.
  - `LEAD-ICB-prime-degree-algebraic-decomposition`: the m = 2 linear Weil-descent ladder, slope 0.647 [0.639, 0.655], 36,440× strong rho at n = 41 (crypto `research/ic_m2_linear_ladder_20261024`).
  - `LEAD-ICE-rootfind-pairtable-oracle`: corrected. Table B rows use `mitm_m2_negfold_walk*` pair tables, not root finding (crypto `docs/ic/leaderboard.json`).
  - Scope notes for the owner criterion of 2026-10-24 and the Option C ruling recorded 2026-10-29 (crypto `research/prime_ic_program_20261029/DECISIONS.md`). These apply to the rootfind, generic-table and constant-engineering leads.
- **Other families.** These folds attach evidence and change no disposition:
  - CZLIFT outcomes, in the three p-adic and global-lift TRF leads.
  - Sheriff strength downgrades (CORR-20261026-c94668).
  - PEXT, Prym, NISTSEED, P-256 parallel tracks, isogeny fibers R5/R6, six-class rho, the known-label graph and cubic-gluing records.

family	leads	closed	open, next step	blocked	unexamined
generic (rho, kangaroo, BSGS, pool)	26	10	11	2	3
ic-binary	22	11	10	1	0
ic-engine	19	5	8	5	1
ic-prime	25	12	6	5	2
transfer	21	14	3	2	2
structure	17	14	3	0	0
exotic	19	8	9	2	0

Counts are from `dispositions.json`, which `figures.py` writes from the committed rows. "Closed" includes exhausted at scale.

Prime-field IC after this review

Three lanes remain, as set out in crypto `research/prime_ic_program_20261029/README.md`:

- **Baseline definition.** The prime-field algebraic track has no baseline until the Semaev-naive + msolve reference arm runs (approved; not yet implemented). The eight Table B prime rows pass the validation gate and are generic-track rows under Option C.
- **Exponent.** Only `LEAD-ICP-pkm-tower-algebraic-oracle` at m ≥ 4 can move it. The PKM note §3.7 shows that no oracle beats rho at m ≤ 3. A derived estimate at 256 bits puts m = 4 within reach of rho only if d_F ≤ 4.
- **Constant.** `LEAD-ICP-prime-ic-constant-engineering`: the 20-bit factor-base gain, then the hash-table prefetch (owner-approved run, waiting on crypto #2783/#2806).

Limitations

- ecc-challenges was not checked out. Its `REVIEW_STATE.json` pin (`3a71596ba`) is carried over from 2026-10-24 and was not re-examined.

- `--check --strict-paths` reports 7 errors that predate this review. They are paths cited from the unmerged crypto PRs #2783, #2806 and #2807, and they resolve when those PRs merge. Plain `--check` passes.
- Merging `main` into the review branch brought in one lead added by another session (148 → 149) and 23 administrative watched paths: decision records, problem records and CI reports. `REVIEW_STATE.json` keeps the stamp's own tally (148 leads), so those paths appear in the next `--delta`.
- The PDF is rendered with reportlab (`build_pdf.py`), because this container has no Typst or LaTeX engine.

Files

- `dispositions.svg`, `dispositions.json`, `figures.py`: the figure, its counts, and the script that draws both from `knowledge/leads/ecdlp/`.
- `atlas-snapshot.md`: `python3 tools/lead_atlas.py --render` at this review.
- `report.pdf`, `build_pdf.py`: this README as a PDF, with the figure redrawn from `dispositions.json`.